seaborn এর জন্য structured cheatsheet। seaborn হলো matplotlib এর উপর তৈরি, statistical visualization-এর জন্য খুব convenient। আমি তোমার জন্য Pandas + Seaborn context সহ complete summary বানাচ্ছি।

1. Import
import seaborn as sns
import matplotlib.pyplot as plt

Standard convention: sns

Optional for Jupyter:

%matplotlib inline

Set style globally:

sns.set(style="whitegrid") # whitegrid, darkgrid, ticks, white
2. Basic Plotting Functions
Function Description Example
sns.lineplot() Line plot sns.lineplot(x='Age', y='Salary', data=df)
sns.scatterplot() Scatter plot sns.scatterplot(x='Age', y='Salary', data=df, hue='Dept')
sns.barplot() Bar chart (with aggregation) sns.barplot(x='Dept', y='Salary', data=df)
sns.countplot() Count of categorical values sns.countplot(x='Dept', data=df)
sns.boxplot() Box plot sns.boxplot(x='Dept', y='Salary', data=df)
sns.violinplot() Violin plot sns.violinplot(x='Dept', y='Salary', data=df)
sns.histplot() Histogram (with KDE) sns.histplot(df['Salary'], bins=20, kde=True)
sns.kdeplot() Kernel Density Estimate sns.kdeplot(df['Salary'], shade=True)
sns.heatmap() Correlation or 2D data sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
sns.pairplot() Pairwise plots sns.pairplot(df, hue='Dept')
sns.jointplot() 2D plot + marginal histograms sns.jointplot(x='Age', y='Salary', data=df, kind='scatter')
3. Customization
Parameter / Function Description Example
hue Color by category sns.scatterplot(x='x', y='y', hue='Category')
style Marker style by category sns.scatterplot(x='x', y='y', style='Category')
size Marker size by value sns.scatterplot(x='x', y='y', size='Value')
palette Color palette sns.barplot(..., palette='Set2')
order Category order sns.barplot(..., order=['A','B','C'])
ci Confidence interval (for bar/line) sns.barplot(..., ci=95)
orient Orientation ('v' or 'h') sns.boxplot(x='Salary', y='Dept', orient='h')
annot Show values in heatmap sns.heatmap(..., annot=True)
linewidth Line width in plots sns.boxplot(..., linewidth=2)
4. Figure & Axes Control
plt.figure(figsize=(10,5)) # global figure size
sns.set_context("talk") # paper, notebook, talk, poster
sns.set_style("whitegrid") # whitegrid, darkgrid, ticks, white

Axes-level vs Figure-level plots:

Axes-level: sns.scatterplot, sns.boxplot, sns.barplot → works with single ax

Figure-level: sns.catplot, sns.pairplot → manages multiple plots/figures

5. Categorical Plots
Function Description
sns.catplot() Figure-level categorical plot (kind='box', 'violin', 'bar', 'strip', 'swarm')
sns.stripplot() Scatter of categorical variable
sns.swarmplot() Non-overlapping scatter of categorical variable
sns.pointplot() Point plot (mean + CI)

Example:

sns.catplot(x='Dept', y='Salary', hue='Gender', kind='bar', data=df)
6. Statistical Plots
Function Description
sns.regplot() Scatter + regression line
sns.lmplot() Figure-level linear regression
sns.residplot() Residual plot for regression
sns.distplot() Histogram + KDE (deprecated, use histplot + kde=True)
sns.kdeplot() Density estimation
sns.jointplot(kind='reg') 2D scatter with regression
7. Matrix / Heatmaps
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title("Correlation Heatmap")
8. Pair & Joint Plots
sns.pairplot(df, hue='Dept', diag_kind='kde')
sns.jointplot(x='Age', y='Salary', data=df, kind='scatter', hue='Dept')
9. Save Plot
plt.savefig("seaborn_plot.png", dpi=300, bbox_inches='tight')
10. Tips & Rules

Seaborn works best with tidy Pandas DataFrames → long format preferred.

hue/style/size → easily encode multiple variables.

Set figure style first → sns.set() or sns.set_style().

For interactive tweaking → ax = sns.plot(...); ax.set(...).

For large datasets → use alpha < 1, smaller markers, and KDEs instead of scatter.